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Research Statement

Overview. I study how people interact with AI and decision-support systems in high-stakes settings, and how to design those systems so that people can learn from them and rely on them appropriately. My work combines controlled experiments with interactive systems, sequential and latent-state models of human behavior, natural language processing (NLP) and large language models (LLMs), and simulation, with applications in health and in the supply chains that deliver essential medicines. A common thread runs through it: the way a system presents information, and the context in which a person receives it, shape what that person does next — and the same system can help some users while harming others. Designing AI training and feedback systems that adapt to their users and are evaluated rigorously in practice is a natural next step for this agenda.

Experiments with Interactive Systems

In my job market paper, a first-authored manuscript under review at Production and Operations Management, I designed a controlled experiment in which 178 participants managed a wholesaler in a gamified, multi-echelon pharmaceutical supply chain during a drug shortage. The game-based platform was developed with collaborators in game design and human–computer interaction, and each week it showed participants a system-recommended order while the information it displayed about the disruption varied across four conditions. Forward-looking capacity information kept decisions close to the recommendation, while information emphasizing the length of the shortage drove large, risky departures from it. Modeling each participant's decision sequence with hidden Markov models and clustering the decoded sequences, I found that the most consequential behavior was concentrated in a small subgroup of users — evidence that interactive systems should be evaluated on how different users respond over time, not only on averages, and that systems which adapt to a user's current state could help those who need it most.

In first-authored work at the 2025 Winter Simulation Conference, I calibrated simulated agents from these data and validated them against real human behavior using distributional tests. The agents reproduced human behavior in most conditions but failed for the most volatile users, revealing where fixed behavioral types break down. This experience with building simulated decision makers and testing their fidelity against real people bears directly on designing simulated personas for training. This research was conducted within a $750,000 NSF-funded project.

NLP and LLMs for Health

As a co-researcher with the FARIL Lab at Mass General Brigham and Harvard Medical School, I led the NLP work in a study published in JMIR Medical Informatics, designing the methodology, building the models, and producing the results; our models identified up to 98% of fall events documented in clinical notes, compared with 26% using diagnosis codes. I then built a context-aware framework, applied to 26,975 emergency-department notes, that layers interpretable rules for negation, temporality, and risk-versus-event language on machine-learning classifiers, with error analysis showing where and why automated language understanding fails. In industry, I built an LLM pipeline with few-shot prompting, schema-enforced structured output, and explicit validation. I work in Python, PyTorch, and TensorFlow.

Working Across Contexts and with Partners

Earlier in my career, I studied a congested hospital emergency department in Iran, where I redesigned its layout and staffing with simulation and optimization, reducing waiting times for the most critical patients; I also conducted a cross-cultural study of social-media discourse on women's health in four

countries. These projects taught me to adapt research to local constraints and to work closely with clinicians and practitioners, which I see as essential for field studies with health and education partners.

Future Research Agenda

1. Learner-adaptive conversational training. A key question for AI role-play and feedback systems is when adapting to the learner improves learning. Building on my latent-state models, I would infer a learner's proficiency and engagement from interaction logs and use it to adapt the difficulty, language, and behavior of a conversational agent. A first project would compare adaptive and non-adaptive versions of a communication-training agent for frontline health workers, testing for whom and when adaptation helps.

2. Controllable, validated personas and feedback. Simulated patients and personas are only useful if they behave realistically and consistently. Extending my agent-validation work, I would develop methods to measure persona fidelity and controllability — comparing simulated and real interactions with distributional tests and error analysis — and evaluate whether automated feedback is accurate enough to act on, using the context-aware error analysis from my clinical NLP work.

3. Field evaluation and heterogeneity. I would help move systems from pilot studies to rigorous field evaluations, including randomized trials where appropriate, measuring learning, sustained use, and transfer of skills. My experience modeling heterogeneous behavior would support analyses of which learners benefit most and why, and I would combine quantitative outcomes with partners' qualitative insight to understand how systems are used in practice.

Timeline. In the first year, I would aim to co-lead the design and pilot of an adaptive training study with health partners and complete a persona-fidelity evaluation framework; in the second year, to contribute to a larger field trial and prepare papers for venues such as CHI, CSCW, ACL and EMNLP, and the Journal of the American Medical Informatics Association, while developing an independent agenda on adaptive, reliable AI training systems for health. I would also be glad to mentor students and contribute to research proposals.